

Image Restoration & Object Recognition under Illumination Extremes

Man Namgung¹, Nitik Jain¹, Patrick Li¹, and Sahen Juneja¹

¹Johns Hopkins University

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Abstract

We evaluate four approaches for correcting overexposed, underexposed, and glare-affected images: BPDHE, a SIFT-based multi-exposure fusion method, and two deep-learning models (U-Net and ResUNet). Classical methods improve global visibility but suffer from artifacts under complex illumination, while deep models achieve higher perceptual and structural fidelity. ResUNet attains the strongest quantitative performance, and U-Net produces the sharpest full-resolution outputs. Our results highlight the limitations of RGB-only enhancement and the advantages of data-driven models for robust exposure correction.

1 Introduction

The goal of our project is to investigate the advantages and limitations of different approaches for correcting overexposed, underexposed, and glare-affected images. Early state-of-the-art work in extreme low-light photography focused on RAW-to-RGB reconstruction, most notably the “Learning to See in the Dark” (SID) [1], which leverages sensor-level RAW data to recover high-quality images even under extremely low illumination. While highly effective, these approaches require RAW Bayer inputs, camera-specific calibration, and substantial computational resources, making them impractical for standard RGB-based workflows. More recent RGB-domain enhancement techniques, such as Zero-DCE++ [2], aim to bypass these constraints by operating directly on sRGB images. Although these methods achieve strong performance on moderately low-light or imbalanced-exposure scenes, our experiments show that Zero-DCE++ struggles on severely underex-

posed images where 8-bit RGB inputs inherently lack the sensor information preserved in RAW data. In contrast, deep learning architectures such as U-Net and ResUNet do perform well, but only when trained with very large, diverse datasets—an impractical requirement for many real-world applications.

In this project, we directly compare classical computer-vision algorithms with modern deep learning-based enhancement models across a range of exposure conditions. As of this report, we have completed implementing the classical methods and have found that, surprisingly, these traditional approaches produce robust and often competitive results without requiring large datasets or heavy training pipelines. This highlights an important insight: classical techniques, despite their simplicity, remain powerful and data-efficient baselines for exposure correction, especially in scenarios where deep models fail or are difficult to train effectively.

2 Datasets

For our experiments, we employ publicly available RGB datasets that are specifically designed for exposure correction and image enhancement. These datasets contain incorrectly exposed images along with corresponding ground-truth references, allowing both quantitative and qualitative evaluation.

The UNet and ResUNet deep learning models were trained and evaluated using the Exposure Correction Dataset introduced by Afifi et al. in “Learning Multi-Scale Photo Exposure Correction” [3]. This dataset provides multiple exposure levels for each scene, paired with high-quality reference images, and is widely used for supervised learning of exposure correction. We also use this dataset to benchmark all four enhancement methods included in our study.

To further assess robustness, the classical BPDHE algorithm was evaluated on the TM-DIED dataset (“The Most Difficult Image Enhancement Dataset”) proposed by Vonikakis et al. [4]. TM-DIED contains challenging real-world photographs with severe illumination distortions, including glare, underexposure, and mixed-lighting conditions. This was conducted as an ablation study rather than a direct comparison, with the goal of understanding the algorithm’s practical applicability in real-world scenarios.

We employ the ExDark [5] dataset to evaluate the SIFT-based enhancement process. ExDark consists of naturally captured low-light scenes spanning 12 object categories and includes a wide range of illumination conditions, making it a meaningful dataset for the objectives of our project. By applying our enhancement models to ExDark, we analyze how the algorithms behave under unstructured and challenging conditions that resemble real deployment environments rather than curated exposure datasets.

All datasets consist exclusively of sRGB images, making them compatible with classical algorithms as well as modern RGB-based learning approaches. Although we initially explored the SID dataset for its relevance to extreme low-light reconstruction, its reliance on camera-specific RAW Bayer inputs makes it unsuitable for our RGB-only pipeline. We therefore reference SID only in our analysis to highlight the limitations of 8-bit RGB data and to motivate why RAW-based reconstruction models cannot be directly compared with RGB enhancement techniques.

3 Evaluation Metrics

To evaluate the performance of the enhancement methods, we rely on three standard reference-based metrics commonly used in image restoration literature.

1. **PSNR (Peak Signal-to-Noise Ratio)** [6]: measures pixel-wise reconstruction fidelity relative to the ground truth. Higher PSNR indicates lower distortion. We use PSNR for quantitative evaluation on paired datasets such as the Exposure Correction dataset.
2. **SSIM (Structural Similarity Index Measure)** [7]: evaluates structural similarity by comparing luminance, contrast, and texture. SSIM is more perceptually aligned with human vision than PSNR and is used on all paired datasets.

3. **LPIPS (Learned Perceptual Image Patch Similarity)** [8]: computes perceptual similarity using deep feature distances from pretrained neural networks. LPIPS is particularly informative for assessing perceptual realism in enhanced images.

4 Methods

We evaluate four complementary approaches to exposure correction and low-light image enhancement, spanning classical computer vision, multi-exposure fusion, and modern deep learning. Each method represents a different class of algorithms, allowing us to analyze their strengths, limitations, and practical applicability across real-world illumination conditions.

4.1 Brightness Preserving Dynamic Histogram Equalization (BPDHE)

BPDHE is a classical contrast-enhancement algorithm that operates in the intensity domain while maintaining global brightness statistics. The method smooths the histogram, identifies dominant peaks, and partitions the histogram into sub-regions. Each sub-histogram is then equalized independently and remapped to a dynamically allocated output range. A final normalization step preserves mean brightness. BPDHE is lightweight and interpretable but can introduce artifacts under complex lighting and often fails on severe under/overexposure.

4.2 SIFT-Based Image Enhancement

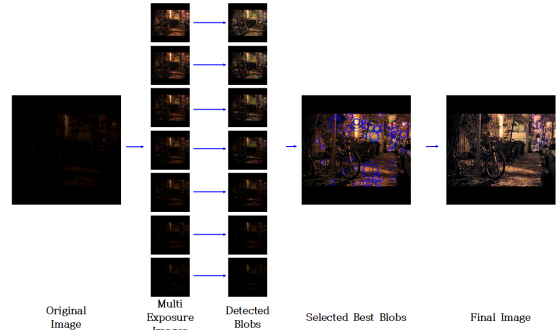


Figure 1: Overview of the SIFT-based multi-exposure fusion pipeline.

To overcome the limitations of single-exposure enhancement, we implement a SIFT-based multi-exposure fusion pipeline. Given an input

image, we synthesize seven exposures (original, three brighter, three darker). SIFT keypoints are extracted and evaluated using a Laplacian-based sharpness measure with contrast, brightness, and saturation normalization. For each keypoint, the best-exposed patch is selected according to the sharpness score:

$$S = 1.3 [(\sigma_L^2 + 0.8\omega) w_b w_s + 0.2 H] \quad (1)$$

$$\begin{aligned} \sigma_L^2 &= \text{Var}(\Delta g), \quad \omega = E(g_x^2 + g_y^2), \\ H &= - \sum_k p_k \log p_k, \\ p_b &= \min\left(\frac{|m - 128|}{128}, 1\right), \\ w_b &= \max(0, 1 - 0.2p_b), \\ s &= \text{mean}(g \leq 5 \text{ or } g \geq 250), \\ w_s &= \max(0, 1 - 0.4s). \end{aligned} \quad (2)$$

Each exposure is also divided into a 20×20 grid with overlap, where the cell with the highest Sobel-gradient magnitude is chosen. Patch-level and grid-level selections are fused using distance-weighted blending, producing a spatially consistent enhanced image. This method effectively handles locally varying illumination but may amplify noise or introduce seams if synthetic exposures are extreme.

4.3 ResUNet for Exposure Correction

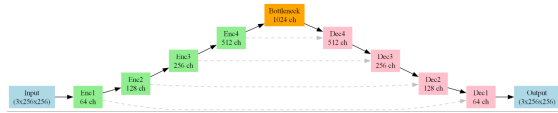


Figure 2: Architecture of the lightweight U-Net/ResUNet model used in our experiments.

We implement a ResUNet architecture trained on the Exposure Correction dataset to evaluate supervised deep learning for RGB enhancement. The model consists of a downsampling encoder with residual blocks, a bottleneck capturing global illumination structure, and an upsampling decoder with skip connections.

Training is performed on $\sim 19,000$ images with $\sim 2,000$ for validation. Pixel fidelity is optimized using PSNR-based loss. ResUNet performs well when trained on large datasets but is highly data-dependent; performance degrades substantially when training samples are limited.

4.4 Lightweight U-Net

To analyze the behavior of SID-style reconstruction under RGB constraints, we design a

lightweight U-Net tailored for exposure correction in sRGB images. The model features a three-level encoder-decoder with skip connections and contains $\sim 7.8\text{M}$ parameters.

Given a single poorly exposed RGB image as input, the model outputs an enhanced version using a combination of L_1 loss, SSIM loss, LPIPS perceptual loss, and an optional exposure-consistency regularizer. This architecture balances performance and efficiency: more expressive than classical methods, significantly lighter than transformer-based models, and compatible with standard datasets.

4.5 Zero-DCE++ Baseline

Zero-DCE++ serves as our unsupervised baseline. It estimates pixel-wise enhancement curves without requiring paired supervision. In practice, the method works on moderately underexposed images but struggles with severe low-light scenes, where lost dynamic range cannot be recovered from sRGB inputs. In our experiments, Zero-DCE++ consistently underperformed compared to supervised models and even classical SIFT-based fusion in extreme illumination cases.

5 Results

5.1 BPDHE Results

5.1.1 Evaluation on TM-DIED Dataset

To evaluate BPDHE on challenging real-world illumination conditions, we tested the method on the TM-DIED dataset. Each input image was first converted to HSV color space to separate luminance from chromatic information. BPDHE was applied only to the value (V) channel, and the enhanced V component was then recombined with the original hue and saturation channels to reconstruct the final output. A full comparison of the performance across all four methods on same images is provided in the Discussion section.

Qualitatively, the results exhibited mixed performance:

- noticeable improvements in overall visibility for moderately underexposed regions,
- but frequent halo artifacts, contrast overstretching, and color shifts in scenes with complex lighting or glare.

This behavior reflects the inherent limitations of histogram-based global enhancement when applied to scenes with strong spatial illumination variations. Detailed evaluation scripts are

provided in the codebase, and a selection of representative results is shown below:



Figure 3: Qualitative results of BPDHE. Top row: enhanced visibility and improved contrast in moderately underexposed outdoor scenes. Bottom row: noise amplification and halo artifacts in extreme low-light scenes with strong light sources, highlighting BPDHE’s limitations under complex illumination.

5.1.2 Evaluation on Afifi Exposure Correction Test Set: BPDHE

On the Afifi test set, we computed three reference-based metrics to quantify reconstruction fidelity against the ground-truth images. The average scores for BPDHE were:

- **PSNR:** 16.149
- **SSIM:** 0.7507
- **LPIPS:** 0.1740

These results indicate that while BPDHE provides some improvement in global brightness and contrast, its pixel-level accuracy and structural consistency remain significantly lower than those of supervised deep learning models. The relatively high LPIPS score further suggests limited perceptual similarity, confirming that BPDHE struggles to recover fine details and natural textures in exposure-corrected outputs.

5.2 SIFT Based Image Enhancement Results

5.2.1 Qualitative Evaluation on ExDark Dataset: SIFT

On the ExDark dataset, the SIFT-based enhancement method successfully recovers local details by selecting the sharpest blob across the set of synthetically generated exposures, enabling the fused output to reveal structural features that are not visible in the original low-light input. The independently constructed base image further improves global tone consistency by stabilizing illumination across regions with varying brightness levels. Adjusting the minimum and maximum pixel-area thresholds allows the pipeline to extract a larger number of fine-grained features, producing more detailed reconstructions when appropriate.

Despite these strengths, several limitations are observed. In areas with low gradient variation—such as sky regions, smooth walls, or other texture-deficient surfaces—keypoint sparsity results in blocky or inconsistent regions in the base image. These artifacts highlight the dependence of feature-driven fusion techniques on local structural richness, and their reduced effectiveness in homogeneous or low-texture areas.



Figure 4: Representative qualitative results of SIFT-based enhancement on the ExDark dataset.

5.2.2 Evaluation on Afifi Exposure Correction Test Set: SIFT

To benchmark the SIFT-based pipeline under supervised conditions, we evaluate it on the Afifi Exposure Correction test set using PSNR, SSIM,

and LPIPS. These metrics quantify pixel fidelity, structural similarity, and perceptual quality, respectively.

The average scores for the SIFT-based image enhancement method are:

- **PSNR:** 12.6438
- **SSIM:** 0.6230
- **LPIPS:** 0.2752

These results indicate that while SIFT-based enhancement can reveal additional structural detail in low-light scenes, it struggles to match the quantitative performance of supervised deep learning models on paired datasets.

6 U-Net Results

6.0.1 Evaluation on Afifi Exposure Correction Test Set: U-Net

The lightweight U-Net model trained for RGB exposure correction demonstrates strong overall performance, achieving substantial improvements over baseline methods such as Zero-DCE++. Qualitative results (Fig. 5) show that the network produces smooth, artifact-free enhancements while preserving structural detail across a range of overexposed and underexposed scenes. The model effectively normalizes global illumination and recovers fine textures, even when trained for a relatively short duration on a modest dataset.

Quantitatively, the U-Net achieves the following average scores on the Afifi test set:

- **PSNR:** 18.44
- **SSIM:** 0.782
- **LPIPS:** 0.209

These results indicate that the U-Net substantially outperforms Zero-DCE++, providing a gain of approximately +6.95 dB in PSNR, along with notable improvements in SSIM and LPIPS. The network also avoids the artifacts commonly seen in classical enhancement methods, offering more stable tonal adjustments and improved perceptual quality.



Figure 5: Qualitative results of the lightweight U-Net on the Afifi Exposure Correction test set. Left: input image; center: U-Net output; right: ground truth.

Although U-Net does not reach the performance levels of RAW-based enhancement models such as SID (which achieve PSNR values in the range of 27–28), this is expected given the restricted dynamic range and nonlinear characteristics of sRGB images. Despite these limitations, the lightweight U-Net provides strong and consistent enhancement results while remaining highly efficient, requiring only ~ 7.8 M parameters and training in approximately two hours on a standard GPU setup. This positions it as a practical middle ground between classical methods, heavy transformer-based architectures, and computationally expensive RAW-domain reconstruction pipelines.

7 ResUNet Results

7.1 Evaluation on Afifi Exposure Correction Test Set: ResUNet

The ResUNet model demonstrates strong performance on the Afifi Exposure Correction test set, producing consistent and artifact-free enhancements across both overexposed and underexposed input images. Compared to classical methods, ResUNet yields noticeably cleaner outputs, with improved global tone and preserved structural detail. Sample qualitative results are shown in Fig. 6, where each row displays the input image (left), the ResUNet prediction (center), and the ground-truth reference (right).

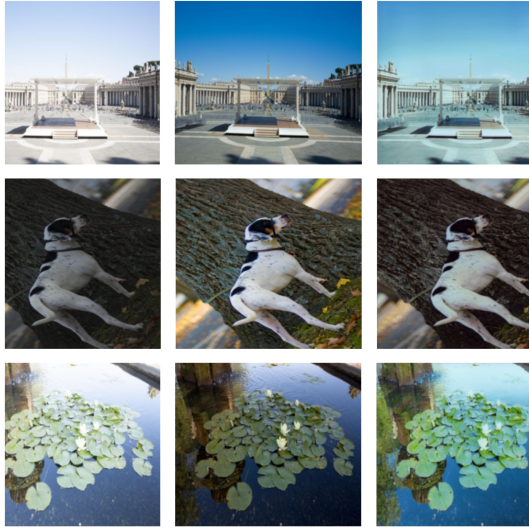


Figure 6: Qualitative results of ResUNet on the Afifi Exposure Correction test set. Left: input image; center: ResUNet output; right: ground-truth reference.

Quantitatively, the model achieves the following average scores:

- **PSNR:** 19.27
- **SSIM:** 0.8433
- **LPIPS:** 0.1171

These results place ResUNet among the highest-performing methods in our benchmark. The best-performing checkpoint (Model 3 at epoch 4) produced the scores listed above. Although the network was trained for a relatively short duration and with limited computational resources, it outperformed classical approaches and avoided the halo or contrast-stretching artifacts often observed in histogram-based and fusion-based methods. However, we note that some outputs remain only partially normalized, exhibiting subtler improvements than larger, more heavily trained models.

In comparison to RAW-based methods such as SID, which report PSNR values in the range of 27–28, our RGB-only ResUNet naturally achieves lower absolute scores due to the restricted dynamic range and loss of sensor information inherent in sRGB inputs. Even so, the model performs competitively within the RGB-only enhancement domain while remaining significantly lighter and faster to train.

8 Cross-Method Evaluation and Analysis

Across all four methods, our experiments highlight a clear trade-off between classical, feature-driven, and deep-learning-based exposure correction approaches. Classical methods such as BPDHE and SIFT excel in specific scenarios but struggle to generalize, whereas the U-Net and ResUNet models consistently deliver higher perceptual and structural fidelity.



Figure 7: Qualitative comparison across all methods. Top to bottom: Input, Ground Truth, SIFT, BPDHE, ResUNet, and U-Net. ResUNet outputs are limited to 255×255 resolution (pixelation visible), while U-Net produces sharp full-resolution predictions.

BPDHE succeeds in moderately low-light scenes by boosting global contrast but fails under complex illumination, producing haloing and color distortions. The SIFT-based approach performs well in structurally rich regions where keypoints are abundant, recovering details that histogram-based methods cannot; however, it breaks down in smooth or texture-poor areas, leading to block artifacts. In contrast, ResUNet and U-Net provide stable, artifact-free reconstructions; ResUNet offers stronger metrics overall but generates 255×255 outputs due to GPU constraints, resulting in mild pixelation when upsampled. The lightweight U-Net, despite being simpler, produces sharper full-resolution images and more consistent global illumination.

A quantitative comparison of all methods on the Afifi Exposure Correction test set is provided in Table 1. These results illustrate the significant performance gap between classical algo-

rithms and modern deep-learning pipelines, particularly in perceptual similarity (LPIPS) and structural coherence (SSIM).

Table 1: Comparison of average metrics for all four methods on the Afifi Exposure Correction test set.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BPDHE	16.15	0.751	0.174
SIFT	12.64	0.623	0.275
U-Net	18.44	0.782	0.209
ResUNet	19.27	0.843	0.117
SID on RGB	17.40	0.554	N/A
SID on RAW	28.88	0.787	N/A

A qualitative comparison is shown in Fig. 7, where the top rows depict the input and ground truth, followed by BPDHE, SIFT, ResUNet, and U-Net. BPDHE enhances global brightness but introduces artifacts; SIFT improves local details but suffers from block inconsistencies; ResUNet generates strong reconstructions but exhibits pixelation due to its smaller output resolution; and U-Net provides the sharpest, most visually consistent enhancement across all conditions evaluated.

Conclusions

In this work, we investigated the strengths and limitations of classical, feature-based, and deep-learning approaches for correcting overexposed, underexposed, and glare-affected images. Classical methods such as BPDHE and SIFT offer lightweight, interpretable pipelines and can improve visibility in specific scenarios, but they struggle with complex illumination and fail to generalize across diverse scenes. In contrast, deep-learning models—particularly U-Net and ResUNet—consistently achieved higher perceptual and structural fidelity, with ResUNet delivering the strongest quantitative performance and U-Net producing sharper full-resolution outputs despite significantly lower computational cost.

Our comparisons highlight that while RGB-domain enhancement is fundamentally constrained compared to RAW-based reconstruction, carefully designed neural architectures can bridge much of this gap and outperform traditional algorithms by a wide margin. Overall, the results emphasize the importance of spatial modeling, perceptual supervision, and data-driven learning for robust exposure correction, pointing toward hybrid or transformer-based approaches as promising directions for future work.

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- **Patrick Li** — implementation of the BPDHE method
- **Man Namgung** — development of the SIFT-based enhancement algorithm from scratch
- **Sahen** — implementation of ResUNet and evaluation of all methods except U-Net
- **Nitik Jain** — implementation of the U-Net pipeline and formalization of the project report

Our final results reflect the combined effort of these individual contributions. The codebase and more updates can be found on the webpage: extreme-illumination-image-restoration.github.io

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